SIEM Project Report: SSH Brute Force Detection using Splunk

1. Project Title

Detection of SSH Brute Force Attacks using Splunk SIEM

2. Objective

The objective of this project is to simulate SSH brute-force login attempts and detect them using Splunk by analyzing system authentication logs. This helps in understanding real-world security monitoring and threat detection techniques used in SOC environments.

3. Tools & Technologies Used

- Splunk Enterprise (SIEM Tool)
- Ubuntu Linux (Log Source Machine)
- Kali Linux (Attacker Machine)
- Hydra (Brute-force tool)
- SSH Service
- VirtualBox (Virtualization)

4. Project Architecture

- Ubuntu system generates authentication logs (/var/log/auth.log)
- Splunk monitors and ingests these logs
- Kali Linux simulates brute-force attack using Hydra
- Splunk detects failed login attempts and identifies attacker IP

5. Implementation Steps

Step 1: Install and Start Splunk

- Installed Splunk Enterprise on Ubuntu

- Started Splunk service using:

./splunk start

- Accessed Splunk Web Interface at:

http://localhost:8000

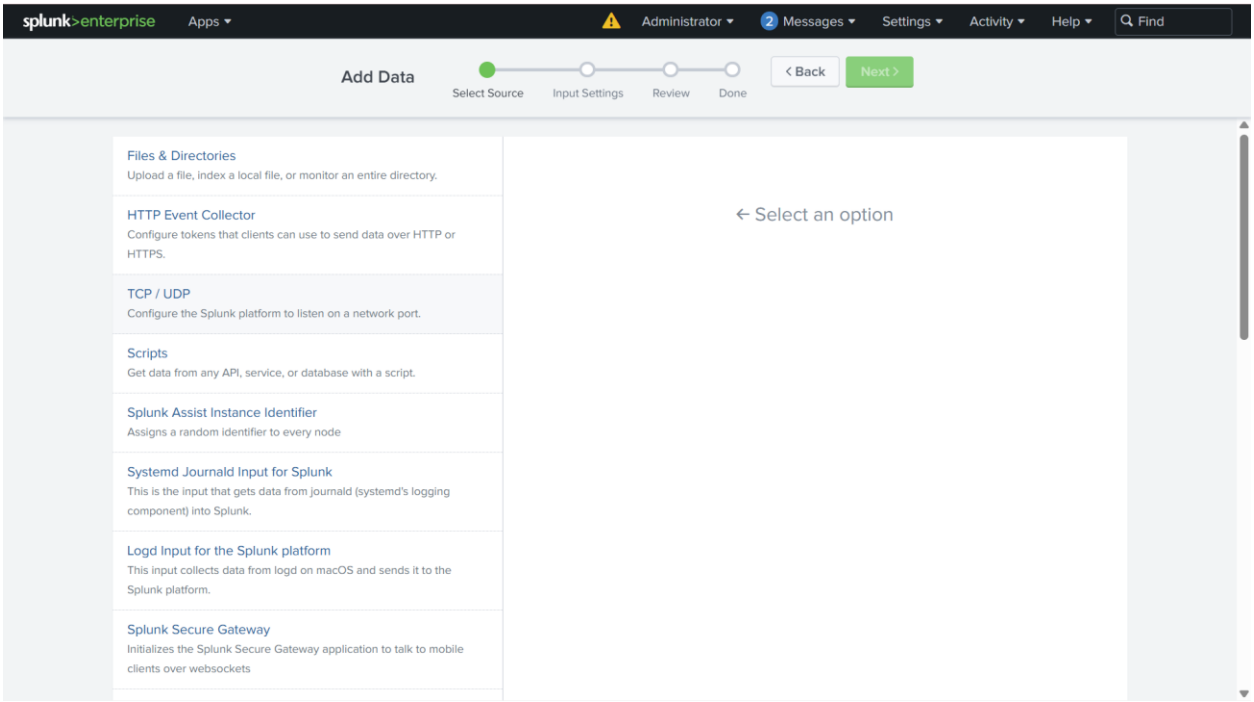
Step 2: Add Log Data to Splunk

- Navigated to **Add Data → Files & Directories**
- Selected log file:

/var/log/auth.log

- Set source type as:

linux_secure

- Indexed logs into default index
- 

Step 3: Enable SSH Service

- Installed and started SSH service:

sudo apt install openssh-server
sudo systemctl start ssh

- Verified service is running on port 22

Step 4: Simulate Brute Force Attack

- Used Kali Linux to perform SSH brute-force:

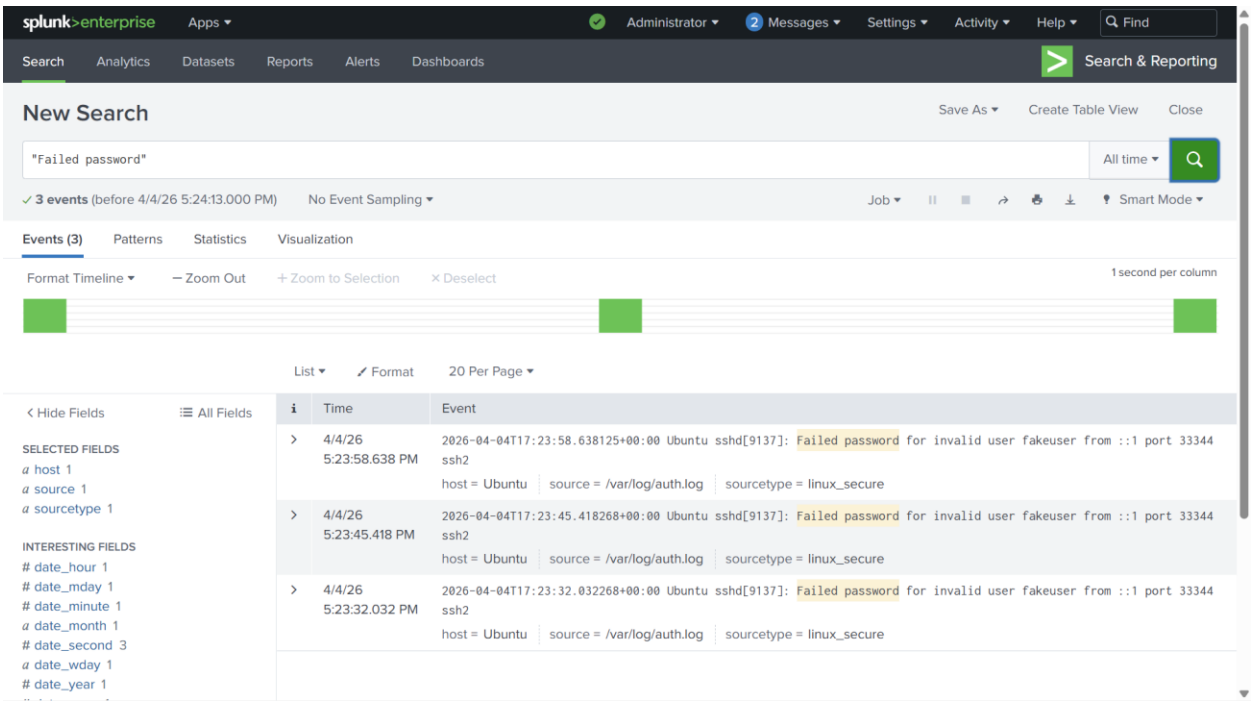
hydra -l msfadmin -P /usr/share/wordlists/rockyou.txt ssh://<target-ip>

- Generated multiple failed login attempts

Step 5: Search Logs in Splunk

- Queried failed login attempts:

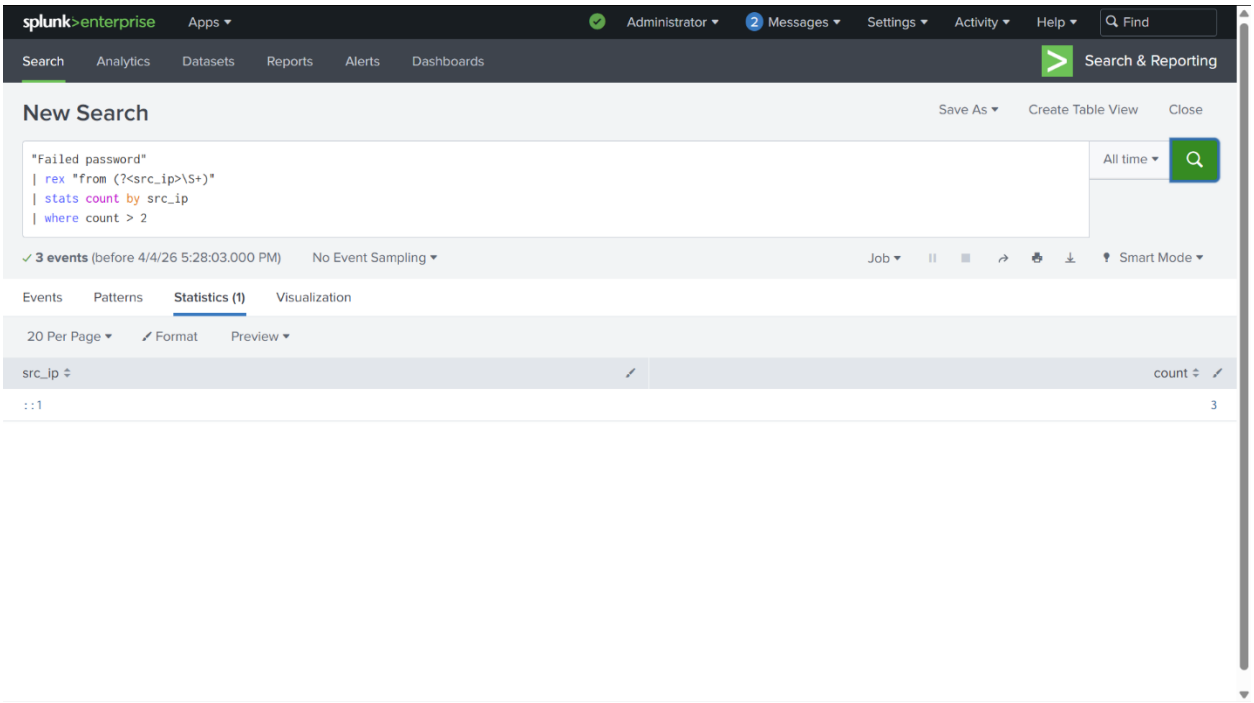
"Failed password"

- Verified logs were successfully ingested
- 

Step 6: Detection Rule Creation

- Extracted attacker IP and counted attempts:

```
"Failed password"
| rex "from (?<src_ip>\S+)"
| stats count by src_ip
| where count > 2
```



6. Results

- Successfully detected multiple failed SSH login attempts from system logs
- Extracted and identified the source IP address using Splunk field extraction
- The detected IP address (::1) represents the local host, indicating that the attack simulation was performed from the same machine
- Created a detection query to highlight repeated failed login attempts (possible brute-force activity)
- Demonstrated real-time log monitoring and analysis using Splunk SIEM

7. Key Learnings

- Understanding of SIEM tools and log analysis
- Hands-on experience with Splunk data ingestion
- Practical knowledge of SSH attack simulation

- Ability to write detection queries (SPL)
- Basics of SOC-level threat detection

8. Conclusion

This project demonstrates how Splunk can be used effectively as a SIEM tool to detect brute-force attacks by analyzing system logs. It provides practical exposure to real-world cybersecurity monitoring and enhances skills required for SOC analyst roles.